

Dual Entropy Gradients and the Tholonic Void-Plenum Equivalence

J. W. Milton

Clarity Coalition

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Abstract

The Tholonic N-D-C framework, when applied at the cosmological scale, predicts two opposing entropy gradients operating simultaneously and structurally. The material domain, identified with the C (Contribution/Integration) role at the cosmic scale, undergoes entropy increase: it disperses, differentiates, and moves from minimum entropy (the material singularity) toward maximum entropy (maximum diffusion, heat death). The conceptual domain, identified with the D (Definition/Limitation) role at the cosmic scale, undergoes entropy decrease: it organizes, becomes more specific, more coherent, and more structured over time, moving from maximum entropy (the conceptual void: one concept, zero differentiation, zero organization) toward minimum entropy (maximum structure, maximum connection, maximum coherence).

This duality is not an additional postulate of the tholonic model. It is a structural consequence of the N-D-C role assignment: C is the dispersive, expansive, accumulating pole, and D is the constraining, organizing, specifying pole. At the cosmic scale, these roles map directly onto the material and conceptual domains respectively. The N state of the universe at any moment is the balance between these two opposing gradients.

The paper further argues that the starting point of the conceptual expansion (the void: a zero-dimensional point of zero differentiation) and its endpoint (infinite concepts organized into infinite mutual connections) are topologically and informationally equivalent. Both are states of zero Shannon entropy in different senses: the void is maximally underdetermined (all possibilities equally probable), while the plenum is maximally determined (all possibilities fully realized). This equivalence is supported by Kuiper's theorem for infinite-dimensional Hilbert spaces, the self-duality of projective geometry in the limit, and the behavior of Shannon entropy at its two extremes.

The paper is explicitly exploratory. The structural argument (dual entropy gradients as a consequence of N-D-C role assignment) is derived from the established tholonic framework. The cosmological claims (void-plenum equivalence, the material/conceptual mapping) are argued structurally and by mathematical analogy, not

by formal proof. Open questions identify what would be required to elevate the central claims from structural argument to theorem.

1 Introduction

The second law of thermodynamics states that entropy in an isolated system does not decrease. This is among the best-confirmed principles in physics. It is also, on inspection, a statement about a specific domain: the material domain, specifically the domain of energy distribution in physical systems.

The question this paper addresses is whether the same principle, or its converse, applies to a different domain: the domain of concepts, ideas, and information. If the material domain is characterized by entropy increase, is the conceptual domain characterized by entropy decrease? And if so, is this duality a consequence of some deeper structural principle, or is it an independent empirical observation?

The Tholonic N-D-C framework [Mil24c] provides a structural answer. The framework assigns three irreducible functional roles to any self-organizing recursive system: N (Negotiation: the emergent stable state), D (Definition/Limitation: the constraining, organizing, specifying role), and C (Contribution/Integration: the expansive, accumulating, dispersive role). At the cosmic scale, the material domain exhibits the properties of the C role: it disperses, expands, and differentiates away from an initial concentration. The conceptual domain exhibits the properties of the D role: it constrains, organizes, and becomes more specifically structured over time. The dual entropy gradients are therefore not an independent observation but a consequence of role assignment.

This paper also argues that the two endpoints of the conceptual entropy trajectory are equivalent to each other and to the single starting point of the material trajectory. The material universe begins as a point singularity (minimum volume, minimum material entropy) and expands toward maximum material entropy. The conceptual universe begins as the void (zero concepts, maximum conceptual entropy) and organizes toward minimum conceptual entropy. The endpoint of full conceptual organization (infinite concepts connected to infinite concepts) is, by several mathematical arguments, equivalent to the starting void. The tholonic recursive structure both predicts and explains this equivalence: at the largest scale, the child N of the entire hierarchy must be equivalent to the parent N that generated it.

The connections to Newton’s “active principles” [New17] and to Sidis’s animate/inanimate distinction [Sid25] are treated in Paper 12 [Mil26a]. The present paper takes the structural argument further and examines its cosmological implications independently.

What this paper provides. A structural derivation of dual entropy gradients from the N-D-C role assignment. A mathematical argument for the void-plenum equivalence via Kuiper’s theorem, Shannon entropy, and projective duality. A set of tholonic predictions about the relationship between material and conceptual domains. Explicit open questions separating what is argued from what remains to be proved.

What this paper does not provide. A formal proof of void-plenum equivalence. A physical model of how the conceptual domain interacts with the material domain. A cosmological model making specific quantitative predictions about entropy values or timescales. The paper is structural and exploratory.

Organization. Section 2 reviews the N-D-C role assignment and the cosmic-scale mapping. Section 3 argues the case for material entropy increase as C expansion. Section 4 argues the case for conceptual entropy decrease as D organization. Section 5 examines Newton’s active principles and Sidis’s animate matter as the interface between the two domains. Section 6 develops the void-plenum equivalence argument. Section 7 provides the mathematical grounding. Section 8 states tholonic predictions and their status. Section 9 poses open questions. Section 10 concludes.

2 The N-D-C Role Assignment at the Cosmic Scale

The tholonic triad assigns three functional roles to any recursive self-organizing system:

- *N* (Negotiation): the emergent stable state. It is not directly measured; it arises from the interaction of D and C. It is simultaneously the product of one level and the source of the next.
- *D* (Definition/Limitation): the constraining, bounding, specifying role. D governs what a system IS: its boundaries, identity, and organization. D reduces possibility to actuality by specification.
- *C* (Contribution/Integration): the accumulating, expansive, dispersive role. C governs what a system DOES: its outputs, relationships, and extensions. C increases the scope of actualized states.

The fundamental stability principle is $D \approx C$: systems are most stable when their constraining and expansive poles are balanced. Imbalance in either direction increases energy cost and degrades the N state.

At the cosmic scale, the assignment is:

Role	Cosmic domain	Entropy behavior
<i>C</i>	Material domain: energy, matter, spacetime	Increases (disperses from singularity)
<i>D</i>	Conceptual domain: information, structure, relations	Decreases (organizes from void)
<i>N</i>	The universe as self-organizing system	The balance between the two gradients

This assignment is not arbitrary. The C role is characterized by accumulation and extension: it grows the scope of what exists. Material entropy increase is precisely this: the universe accumulates more and more realized microstates, extending its physical configuration space. The D role is characterized by specification and constraint: it reduces the space of possibilities to a more precisely organized subset. Conceptual entropy decrease is precisely

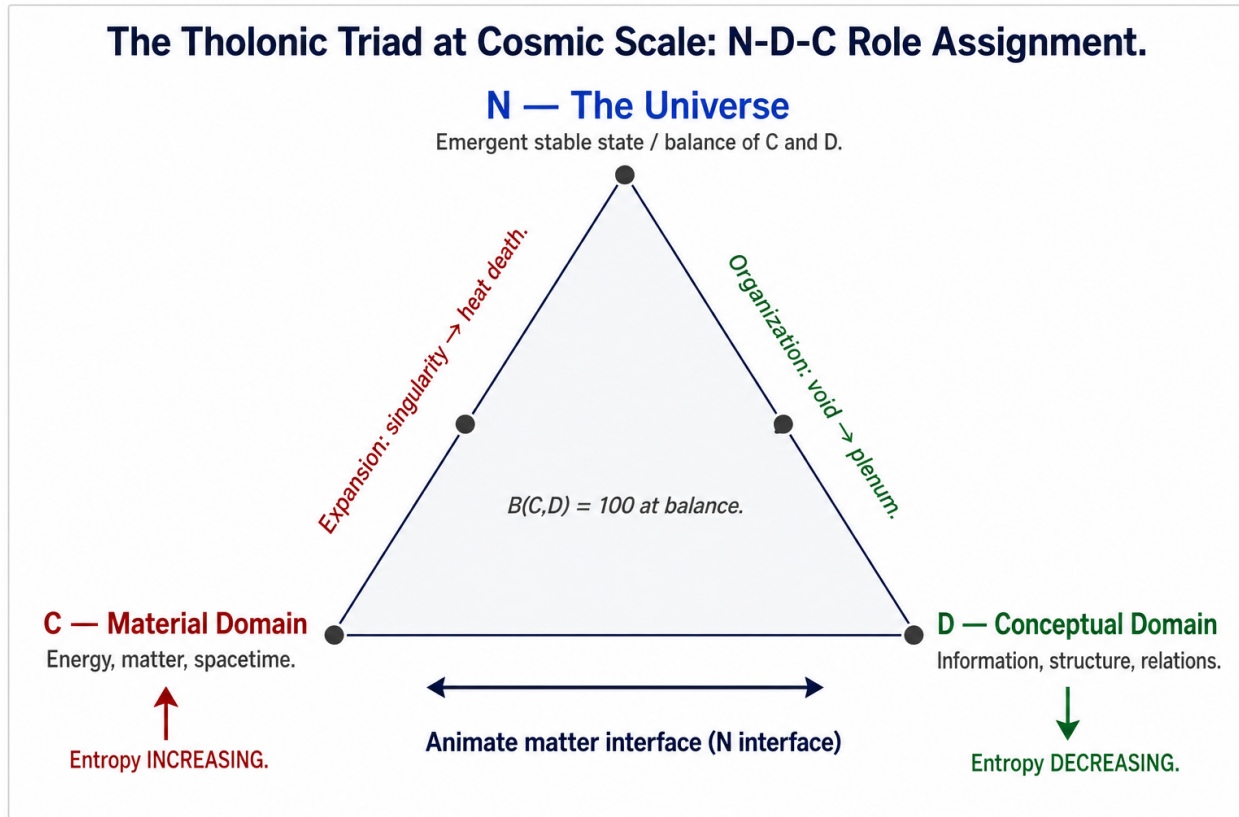


Figure 1: Figure 1: The Tholonic Triad at Cosmic Scale. N (Universe, top vertex) is the emergent balance of D (Conceptual Domain, bottom-left) and C (Material Domain, bottom-right). The bottom edge marks the animate matter interface where the two entropy gradients meet.

this: the space of concepts becomes more organized, more internally constrained, and more specifically structured over time.

Figure 1: The tholonic triad at the cosmic scale. The D and C roles map directly to the conceptual and material domains. The bottom edge, where D and C meet, corresponds to animate matter: the interface at which D-domain entropy decrease achieves material instantiation.

3 Material Entropy Increase as C Expansion

The material universe began, in the standard cosmological model, as a state of extreme concentration: high energy density, minimum spatial extension, minimum material entropy (in the sense of Boltzmann: few accessible microstates). Since the initial singularity, the universe has expanded. Energy has dispersed. Structures have formed and dissolved. The number of accessible microstates has grown. Material entropy has increased and continues to increase.

In tholonic terms, this is the C role operating at the cosmic scale. The C pole of the tholonic triad is the accumulating, extending, contributing pole. It disperses from a concentrated source, increases the scope of what exists, and adds to the running state. The material Big Bang is the initial condition of the C expansion: a maximally concentrated C source beginning its accumulation outward.

The trajectory is:

$$\text{Material singularity (C}_{\min}) \longrightarrow \text{Expansion} \longrightarrow \text{Heat death (C}_{\max})$$

At heat death, material entropy is maximized: all energy is uniformly distributed, no free energy remains, no further differentiation is possible. The C expansion is complete. Every possible material microstate has been realized and dispersed.

This trajectory is monotonic and irreversible in the C direction. There is no material return to the singularity. The tholonic balance principle predicts this: once C has expanded beyond the $D \approx C$ balance point, the system cannot recover the N state without a D input of equal magnitude. In a universe where D is the conceptual domain (non-material), that input cannot act on the material domain directly except through the interface of animate/vegetative systems.

4 Conceptual Entropy Decrease as D Organization

The conceptual domain begins from a different starting point. The simplest possible concept is the void: not the physical vacuum (which has structure, energy, and quantum fluctuations) but the pure conceptual void: zero differentiation, zero relationship, zero organization. Every concept is equally probable; no concept is distinguished from any other. This is maximum

conceptual entropy in the Shannon sense: maximum uncertainty, minimum information, a uniform distribution over all conceptual possibilities.

From this starting point, the conceptual trajectory is the inverse of the material one:

$$\text{Conceptual void (D}_{\text{max entropy}}) \longrightarrow \text{Organization} \longrightarrow \text{Plenum (D}_{\text{min entropy}})$$

Each act of conceptual organization reduces conceptual entropy: it distinguishes one concept from another, establishes a relationship, specifies a constraint. Mathematics, physics, logic, and language are all instruments of this process. They do not discover pre-existing structure passively; they participate in the D-domain process of progressive specification.

The trajectory is driven by the D role’s defining property: specification. D reduces the space of possibilities to a more precisely organized subset. Every theorem proved, every relationship identified, every distinction drawn is a decrease in conceptual entropy. The D expansion is toward minimum conceptual entropy: a state in which every concept is fully specified by its relationships to every other concept.

This trajectory is also monotonic: conceptual entropy decreases over time, in the sense that the organized body of concepts only grows more organized. Individual concepts can be discarded or revised, but the overall degree of organization of the conceptual space increases. What is lost in one branch of inquiry is typically recovered and exceeded in another.

Figure 2: The two opposing entropy trajectories. Material entropy (orange, C domain) increases monotonically from the singularity toward heat death. Conceptual entropy (blue, D domain) decreases monotonically from the void toward the plenum. The crossing point is the N state: the region of local entropy reversal occupied by animate matter and Newton’s active principles.

5 Newton’s Active Principles and Sidis’s Animate Matter

Newton’s unresolved question, posed in Query 31 of the Opticks [New17], was the mechanism of “active principles”: non-mechanical organizing forces that produce coherence, structure, and persistence that mechanical operations cannot generate. In the tholonic reading, active principles are N states. They are the local instantiations of D-domain organization acting within the C-domain material world. A living organism, a crystal, a stable molecular structure: each is a site where the D-domain entropy decrease has achieved material instantiation, temporarily reversing local material entropy against the global C-domain trend.

Sidis [Sid25], approaching the same boundary from statistical mechanics, proposed that animate matter is matter that locally reverses entropy. His argument was that this reversal is statistically necessary given time-symmetric microscopic laws. In the tholonic reading, the statistical necessity argument corresponds to the structural necessity of N emergence: given D and C interacting under the minimality conditions of Paper 3 [Mil24c], N must emerge at some balance point. Animate matter is the material trace of that emergence: the N state made locally visible in the C domain.

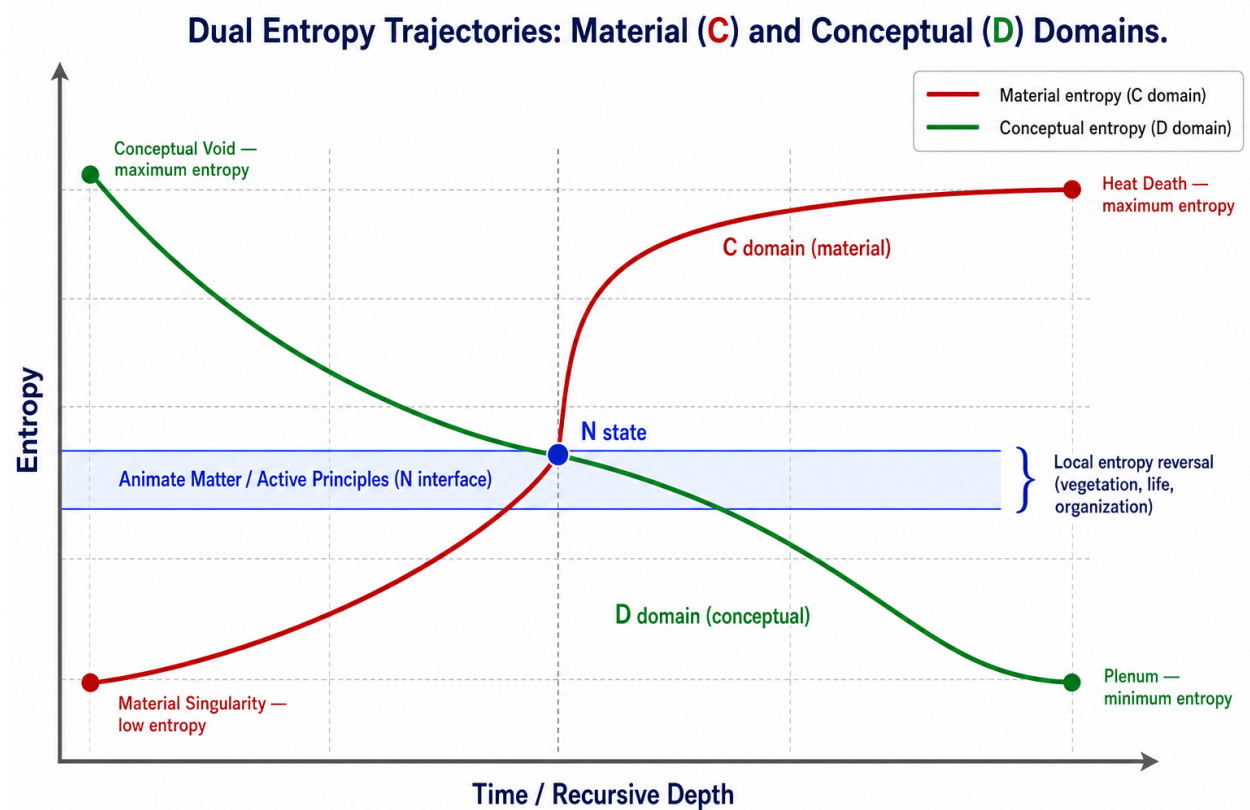


Figure 2: Figure 2: Dual Entropy Trajectories. The material (C) domain begins at minimum entropy and rises toward heat death (orange). The conceptual (D) domain begins at maximum entropy and falls toward the plenum (blue). They cross at the N state, the region where animate matter and active principles operate.

The dual entropy framework extends both Newton’s and Sidis’s observations: they were each identifying the interface between the two domains. Animate matter (Newton’s vegetable spirit, Sidis’s entropy-reversing systems) is not simply a local exception to the second law. It is the point at which the D-domain entropy decrease gradient achieves material expression, temporarily overriding the C-domain entropy increase gradient at the local scale. The interface is the N state.

6 The Void-Plenum Equivalence

The most striking structural claim of this paper is that the starting point of the conceptual trajectory and its endpoint are equivalent. This requires argument.

The conceptual void is the state of maximum conceptual entropy: zero differentiation, zero organization, zero structure. Every possible concept is equally likely. No concept has priority. In Shannon’s terms, this is the uniform distribution over all possibilities: maximum entropy $H = \log_2 N$ where N is the total number of possible concepts (or, in the limit, $H \rightarrow \infty$ as $N \rightarrow \infty$). Geometrically, this is a zero-dimensional point: no distinctions have been drawn, so no dimensions of conceptual space exist.

The conceptual plenum is the state of minimum conceptual entropy: all concepts fully specified by their relationships to all other concepts. Every distinction has been drawn. Every relationship has been established. Every concept is fully determined by its position in the network of all concepts. In Shannon’s terms, this is a degenerate distribution where every concept’s state is fully predictable from the others: entropy $H = 0$. Geometrically, this is an infinite-dimensional space: every possible distinction defines a dimension.

The paradox is that both endpoints have entropy $H = 0$ (or its limiting analogue). The void has zero entropy because there is only one state (undifferentiated nothingness: all possible concepts are the same state, namely “no concept”). The plenum has zero entropy because every concept is fully determined by its connections: zero residual uncertainty. Both are maximally determined, from opposite directions.

This is the tholonic prediction: the parent N at the top of the hierarchy (the void, undifferentiated source) and the child N at the bottom (the plenum, fully realized actuality) are the same entity viewed from opposite ends of the recursive descent. The full tholonic cycle: void (parent N) differentiates into D (conceptual domain) and C (material domain), which interact to produce organized reality (child N), which, when the hierarchy is fully realized, contains all possible N states, and is therefore equivalent to the void that contained all possible N states in potential form.

The distinction between the void and the plenum is the distinction between potential and actual: both are complete, but one is complete in latency and the other in realization. A universe that has realized all possible N states through D-C negotiation is, from the outside, indistinguishable from the void that contained all of them as undifferentiated potential.

7 Mathematical Grounding

Three mathematical results support the void-plenum equivalence. None constitutes a proof of the full claim, but each provides structural evidence that the equivalence is not merely philosophical.

7.1 Kuiper's Theorem

Kuiper's theorem (1965) states that the unitary group $U(H)$ of an infinite-dimensional separable Hilbert space H is contractible in the norm topology. This means $U(H)$ has the homotopy type of a point: all of its homotopy groups vanish. An infinite-dimensional Hilbert space, under its natural unitary symmetry group, has the same topological complexity as a single point.

More precisely: in finite dimensions, $U(n)$ has rich homotopy ($\pi_1(U(n)) = \mathbb{Z}$, etc.). As $n \rightarrow \infty$, the homotopy groups stabilize and then vanish. The topological structure of infinite-dimensional unitary symmetry collapses to the trivial structure of a point. The infinite-dimensional conceptual space, under its natural symmetry operations, is topologically a point.

This is not the same as saying an infinite-dimensional space is a point. It says that under the natural operations defined on it, its invariant structure is point-like. The void (a zero-dimensional point) and the plenum (an infinite-dimensional space with collapsed homotopy) share this invariant.

7.2 Shannon Entropy at Extremes

Shannon entropy is:

$$H(p) = - \sum_i p_i \log_2 p_i$$

It achieves its maximum for a uniform distribution over n states: $H = \log_2 n$. As $n \rightarrow \infty$, $H \rightarrow \infty$. This is the void in one sense: infinite uncertainty, no structure.

It achieves its minimum ($H = 0$) for a degenerate distribution: all probability on one state. This is both the void in another sense (only one possible state: undifferentiated nothingness) and the plenum (only one possible state: fully determined by all relationships).

The symmetry is: the void is $H = 0$ because there are no distinctions ($n = 1$). The plenum is $H = 0$ because all distinctions are fully determined. Between them, H rises through the conceptual expansion (as distinctions are drawn and possibilities proliferate) and then falls back to zero as the system becomes fully organized.

Figure 3: Shannon entropy along the conceptual expansion axis. Both the void ($H = 0$: one undifferentiated state) and the plenum ($H = 0$: every state fully determined) are states of zero entropy, approached from opposite directions. The U-shaped curve is the trajectory of the D domain from Section 4. The void and plenum are the two fixed points of the projective duality discussed in Section 7.3.

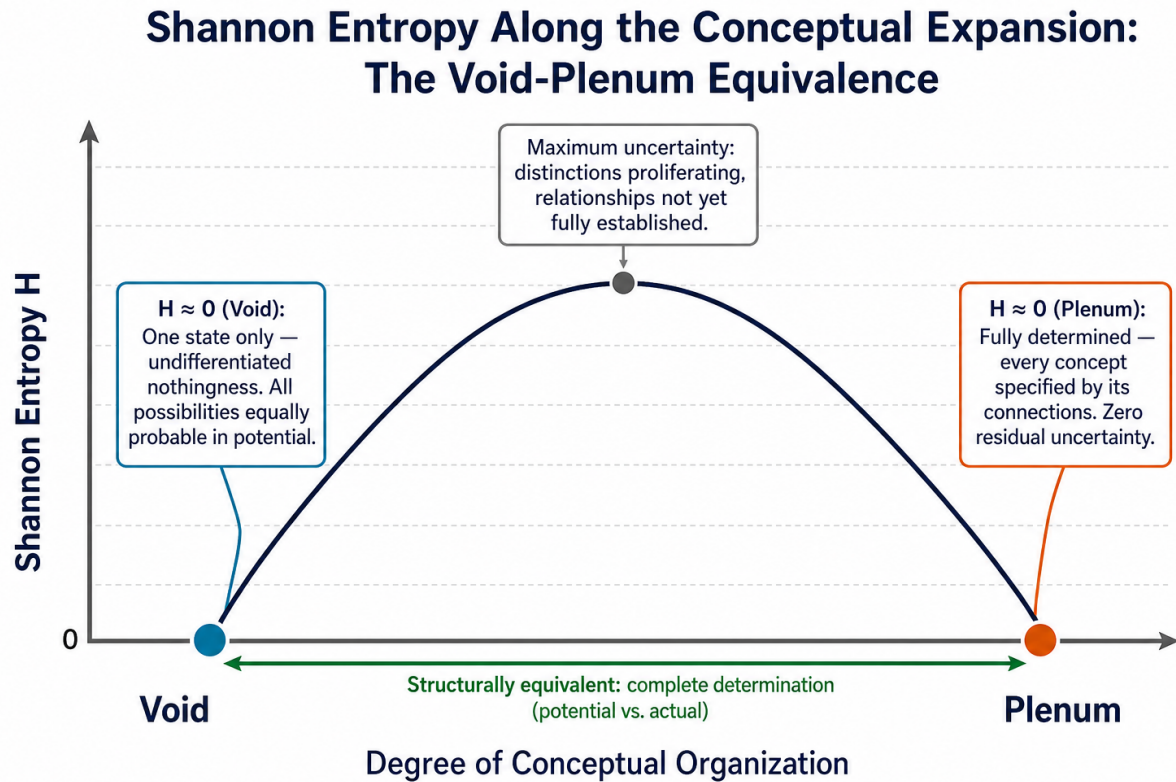


Figure 3: Figure 3: Shannon Entropy Along the Conceptual Expansion. Both the void (left) and the plenum (right) are states of $H=0$, approached from opposite directions. The peak marks maximum uncertainty: distinctions proliferating but not yet organized. The double-headed arrow marks the structural equivalence of the two endpoints.

7.3 Projective Self-Duality

In projective geometry, points and hyperplanes are dual: every theorem about points corresponds to a theorem about hyperplanes under the duality transform. A projective point is “a line through the origin” and a projective hyperplane is “a codimension-1 subspace.” In n dimensions, this duality is exact.

In the infinite-dimensional limit, projective duality becomes self-referential in specific senses. The dual of an infinite-dimensional projective space under certain function-space topologies is homeomorphic to the original space (by the Fréchet-Riesz representation theorem for Hilbert spaces: $H \cong H^*$). The space is its own dual. A fully connected infinite-dimensional conceptual space, where every concept determines a hyperplane of related concepts, is self-dual under the same duality that maps a point to its polar hyperplane.

The void (a single undifferentiated point, no polar hyperplane defined) and the plenum (every point has a polar hyperplane, the space is self-dual) are the two fixed points of this duality operation. They are the points at which the duality map is trivially satisfied: the void because there is nothing to map, the plenum because the map is the identity.

8 Tholonic Predictions

The structural argument produces several predictions about the relationship between the material and conceptual domains. These are stated here with their current status.

Prediction 1: Animate matter marks D-C interface locations. Sites of local material entropy reversal (animate matter, living systems, self-organizing structures) are sites where D-domain organization achieves material instantiation. They should be analyzable as local N states with measurable D-C balance. *Status: testable via the tholonic balance functional $B(D, C)$ applied to biological and self-organizing systems. Supported qualitatively by the vegetation manuscript analysis (Paper 12 [Mil26a], Appendix B).*

Prediction 2: The conceptual entropy of a domain decreases as a function of the number of established relationships per concept. As a field of inquiry matures, the ratio of established relationships to total concepts should increase monotonically. The conceptual entropy of mature fields should be measurably lower than that of emerging fields. *Status: testable via bibliometric or knowledge graph analysis. Not yet tested.*

Prediction 3: The phi-optimal balance point ($D \approx C$) marks the transition between material-entropy-dominated and conceptual-entropy-dominated dynamics. Below the balance point, material entropy increase dominates (C-dominant, dispersive behavior). Above it, conceptual entropy decrease can temporarily dominate (D-dominant, organizing behavior). The 61.8% threshold of the tholonic balance functional marks this transition. *Status: structurally derived from the N-D-C framework. Quantitative test requires identifying measurable proxies for D and C in physical and biological systems.*

Prediction 4: The two endpoints of the full tholonic hierarchy are informationally indistinguishable. A universe that has realized all possible N states through D-C negotiation should be informationally equivalent to the undifferentiated void. *Status: argued*

via Kuiper's theorem and Shannon entropy (Section 7). Not formally proved. This is the most speculative prediction in the paper.

9 Open Questions

9.1 The formal equivalence of statistical and structural necessity. Sidis argues that animate matter is statistically necessary given time-symmetric microscopic laws. The tholonic model argues that N emergence is structurally necessary given D-C interaction under the minimality conditions of Paper 3 [Mil24c]. Are these equivalent claims? If yes, the tholonic framework provides a structural proof of what Sidis could only argue statistically, and the dual entropy gradients follow as a corollary.

9.2 The Landauer bridge. Landauer's principle states that erasing one bit of information dissipates at least $kT \ln 2$ of energy into the material domain. This is a quantitative connection between conceptual operations and material entropy. Does it provide a measurable exchange rate between D-domain entropy decrease and C-domain entropy increase? If the dual gradients are balanced at the N state, the Landauer rate may set the equilibrium condition.

9.3 The starting-point identification. The material singularity and the conceptual void are described as the starting points of the two trajectories. Are they the same starting point? Is the material Big Bang and the conceptual void the same N state viewed through the D and C lenses respectively? If yes, the universe has a single parent N and two diverging trajectories, exactly the tholonic structure.

9.4 Kuiper's theorem as a tholonic statement. Kuiper's theorem says the infinite-dimensional unitary group is contractible. Is this a tholonic statement: that the fully realized N state (the plenum of all unitary transformations) has the same invariant structure as the parent N (the point from which they all derived)? Translating Kuiper's theorem into tholonic grammar may provide the formal bridge between the mathematical result and the structural claim.

9.5 Entropy and the tholonic constants. The five constants produced by the tholonic ladder recurrence ($\pi/4$, φ , e , $\sqrt{2}$, $\ln 2$) all appear in entropy-related expressions: $\ln 2$ in Landauer's principle and binary Shannon entropy, e in Boltzmann's entropy formula, π in the entropy of the Gaussian distribution. Is the appearance of the tholonic constants in entropy formulas a structural consequence of the D-C role assignment, or coincidental?

10 Conclusion

The Tholonic N-D-C framework, applied at the cosmic scale, assigns the material domain to the C role (expansive, accumulating, dispersive) and the conceptual domain to the D role (constraining, organizing, specifying). This assignment predicts dual entropy gradients: material entropy increases while conceptual entropy decreases. The two trajectories begin

from equivalent starting points (the material singularity and the conceptual void are both states of zero differentiation) and converge toward an equivalent endpoint (the plenum of all realized N states is informationally equivalent to the void of all potential N states).

This duality was anticipated by Newton’s distinction between passive mechanical principles and active organizing principles, and by Sidis’s distinction between inanimate (entropy-increasing) and animate (entropy-reversing) matter. Both investigators were identifying the interface between the C and D domains, the N state where material and conceptual entropy gradients meet and locally balance. Neither had the structural framework to explain why the interface exists or why the two gradients are opposed.

The tholonic model provides that framework. The dual entropy gradients are not empirically independent observations. They are the C and D roles operating at the largest observable scale, with the universe itself as the N state of their ongoing negotiation.

The void and the plenum are the same entity. They are the parent N and the fully realized child N of the same recursive tholonic hierarchy, indistinguishable from outside the recursion because both are states of complete determination: one in potential, one in actuality. The universe is the process by which one becomes the other.

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